

Practice set: one idea at a time

Temporal representation and feedforward transformations

Questions. Solutions are in a separate file. Leave that file closed.

How to use this

The practice test puts three decisions inside one question: which formula, which convention, and what the sketch is marking. This set splits those decisions apart, then puts two of them back together at the end.

Each section states one idea, then asks two questions. The first question shows the move. The second changes the numbers, the sign, or the shape, so a memorized picture from the first one will not survive.

1. Read the idea box. Attempt the first question for about ten minutes.
2. Open the solution for that question only. Read the working, close the file, and rewrite the answer on a blank page.
3. Attempt the second question with the solutions closed. If it fails, the gap is the idea, not the arithmetic. Restate the idea box from memory, then try again before you read that solution.
4. When every second question in a section is possible from a blank page, redo Practice Test 2 closed book.

Conventions. Normalized LIF: rest 0, threshold 1, $\tau_{RC} = 20$ ms, $\tau_{\text{ref}} = 2$ ms. Series convention $x(t) = \sum_k X(\omega_k) e^{i\omega_k t}$. Parseval: $\frac{1}{T} \int_0^T x(t)^2 dt = \sum_k |X(\omega_k)|^2$. A spike train is a sum of deltas. Rate-mode activities are in hertz. Decoders use $D^T = (AA^T + N\sigma^2 I)^{-1} AX^T$ when a penalty is present, with A of shape $n \times N$ and X of shape $d \times N$.

Leave exponentials and logarithms symbolic. Label every sketch axis.

1 Hertz and radians per second

Idea for this section

Ordinary frequency f counts cycles per second. Angular frequency is the same information in radians per second: $\omega = 2\pi f$. A real signal has content at $+\omega$ and at $-\omega$.

1. A tone sits at 6 Hz. Give ω in rad/s. A second tone sits at $\omega = 10\pi$ rad/s. Give its frequency in hertz.
2. A signal on $T = 1$ s contains a 4 Hz cosine and a 7 Hz cosine. List every angular frequency at which its spectrum can be nonzero.

2 Bins on a signal of length T

Idea for this section

On an interval of length T the bins are $\Delta f = 1/T$ hertz apart, so $\Delta\omega = 2\pi/T$. Bin number k sits at $\omega_k = k \Delta\omega$. With samples spaced by Δt , there are $N = T/\Delta t$ samples, and the highest representable frequency is the Nyquist frequency $1/(2\Delta t)$.

1. Take $T = 1$ s, $\Delta t = 1$ ms, and a band limit of 4 Hz. The DC bin is set to zero. Find N , $\Delta\omega$, and the Nyquist frequency in hertz. List every ω_k whose coefficient may be nonzero. How many independent real random numbers does conjugate symmetry leave free?
2. Take $T = 0.5$ s and a band limit of 8 Hz, again with DC removed. No time step is given, so there is no Nyquist frequency to compute. Find Δf and $\Delta\omega$, list the nonzero ω_k inside the limit, and count the independent real numbers.

3 What makes $x(t)$ real

Idea for this section

A coefficient $X = a + ib$ contributes $2a \cos \omega t - 2b \sin \omega t$ once it is paired with $X(-\omega) = \overline{X(\omega)}$. That pairing is conjugate symmetry. Copying $X(\omega)$ onto $-\omega$ without conjugating leaves an imaginary part whenever $b \neq 0$.

1. A real signal has $X(\omega) = 3 - i$ at one positive frequency. Write $X(-\omega)$. Write the pair's contribution as a sum of a cosine and a sine.
2. Someone claims a real signal may use $X(\omega) = X(-\omega) = i$. Form $Xe^{i\omega t} + Xe^{-i\omega t}$ and show what goes wrong. Then write the $X(-\omega)$ that actually produces a real signal, and the resulting sine or cosine.

4 One tone, written as coefficients

Idea for this section

Under the series convention, a cosine of amplitude A splits equally onto the two sides: $X(\omega) = X(-\omega) = A/2$. A sine of amplitude A is the pair $X(\omega) = -iA/2$ and $X(-\omega) = iA/2$. Check a proposed pair by rebuilding $2a \cos \omega t - 2b \sin \omega t$.

1. On $T = 1$ s, $x(t) = 6 \cos(2\pi t)$. Find X at every nonzero frequency. Rebuild $x(t)$ from those coefficients as a check.
2. On the same interval, $x(t) = 4 \sin(6\pi t)$. Find the frequency in hertz and the two coefficients. Rebuild the sine from a and b as a check.

5 RMS from the coefficients

Idea for this section

Parseval says the mean square equals the sum of $|X(\omega_k)|^2$ over every bin, including both sides of zero. RMS is the square root of that mean square. A cosine of amplitude A has mean square $A^2/2$, which must agree with Parseval. Multiplying $x(t)$ by a constant multiplies every coefficient by that constant, so the shape of $|X(\omega)|$ is unchanged.

1. Using both Parseval and the $A^2/2$ rule, find the RMS of $x(t) = 6 \cos(2\pi t)$. The two routes should agree.

2. A conjugate-symmetric signal on $T = 1$ s has $X(\pm 2\pi) = 2$ and $X(4\pi) = 1 + i$, with $X(-4\pi)$ the matching conjugate. Find the RMS. By what factor must $x(t)$ be multiplied so that the RMS becomes 1? What happens to the shape of $|X(\omega)|$?

6 Band-limited white noise

Idea for this section

The notebook draws random coefficients inside a frequency limit, conjugates them, zeros the rest, inverse-transforms, and rescales to a requested RMS. “White” means the in-band draws share one distribution, so the average of $|X|$ over many signals is flat up to the limit and zero beyond it. One signal looks ragged. The RMS rescaling changes the height of that plateau and does not move the cutoffs. The horizontal axis of the average plot is ω in rad/s, and a real spectrum has a cutoff on each side of zero.

1. A real, zero-mean random signal has duration 1 s, RMS 1, and a white spectrum band-limited to 8 Hz. Sketch the ensemble-average Fourier magnitude against ω . Label both cutoffs with units. Do not calculate the plateau height.
2. Repeat the sketch for a 3 Hz limit, still at RMS 1. In two sentences, say what is the same about the two sketches, and what changes in the time trace when the limit drops from 8 Hz to 3 Hz with RMS held fixed.

7 Current, before any sketch

Idea for this section

The normalized current is $J = \alpha e x + J^{\text{bias}}$. Between spikes, voltage is pulled toward J . A spike is possible only when that equilibrium is strictly above the threshold, $J > 1$. Compute J for each encoder before thinking about the shape of $v(t)$.

1. Two neurons use $J = e x + 1$ at the constant input $x = 1$, with encoders $e = +1$ and $e = -1$. Find both currents. Which neurons can spike?
2. Two neurons use $J = 2e x + 3/2$ at the constant input $x = -1/2$, again with $e = \pm 1$. Find both currents. Which neurons can spike?

8 Voltage traces

Idea for this section

From $v(0) = 0$ the subthreshold solution is $v(t) = J(1 - e^{-t/\tau_{RC}})$. The curve is steepest at the start and bends toward the horizontal line $v = J$. If it hits 1, the neuron spikes, v resets to 0, and v stays at 0 for τ_{ref} . If $J \leq 1$, there is no spike. If the differential equation would pull v below 0, the simulation clamps v at 0.

1. Using the currents from the first question of the previous section, sketch both voltages against time. Mark the threshold. Mark spikes and refractory intervals wherever they occur. Exact spike times are unnecessary.

- Using the currents from the second question, sketch both voltages the same way. Separately: if a neuron has $J = -1$ and starts at $v = 0$, what does the trace do, and what is the clamp for?

9 Time to reach threshold

Idea for this section

Set $J(1 - e^{-t/\tau_{RC}}) = 1$ and solve, which is valid only for $J > 1$:

$$t_{th} = -\tau_{RC} \ln(1 - 1/J).$$

The inter-spike interval is $\tau_{ref} + t_{th}$. At $J = 1$ the voltage approaches threshold and arrives only at infinite time.

- Take $J = 2$ and $\tau_{RC} = 20$ ms. Find t_{th} and the inter-spike interval. Leave $\ln 2$ in the answer.
- Repeat for $J = 1$ and for $J = 4$. For $J = 1$, say why there is no finite spike time. For $J = 4$, write t_{th} and explain, without a calculator, why it is shorter than the $J = 2$ result.

10 Two spike trains, one $r(t)$

Idea for this section

A symmetric opponent pair has encoders $+1$ and -1 , the same gain, and the same bias. Least squares then gives decoders $d_- = -d_+$. The decoded value collapses to $d_+(a_+ - a_-)$. The signed train $r = a_+ - a_-$ is that combination. If the two neurons are not a matched pair, the two activities have to stay separate.

- Start from $\hat{x} = d_+a_+ + d_-a_-$ and the relation $d_- = -d_+$. Show that $\hat{x} = d_+(a_+ - a_-)$.
- The two neurons now have unrelated decoders $d_+ = 0.4$ and $d_- = 0.1$. Can a single signed train $r = a_+ - a_-$ still carry the decoded value? Write the expression you would actually compute.

11 The optimal filter as a formula

Idea for this section

Decoding $\hat{x} = r * h$ is, at each frequency, multiplication $\hat{X}(\omega) = R(\omega)H(\omega)$. The ensemble-averaged filter that minimizes the power of $X - RH$ is

$$H(\omega) = \frac{\langle X(\omega) \overline{R(\omega)} \rangle}{\langle |R(\omega)|^2 \rangle}.$$

The brackets average over signals. The conjugate sits on R . Outside the input band, $X = 0$ and the leftover response is uncorrelated with the input, so the numerator is zero and $H = 0$.

- Write $H(\omega)$. In one line each, say what X , R , the conjugate, the brackets, and the denominator are. These are the five things the formula is made of.
- A band-limited input is zero past its cutoff, and the response noise there is uncorrelated with the

input. Explain why $H(\omega) = 0$ at those frequencies even though $|R(\omega)|$ is not zero. Then say what that implies for $\hat{X}(\omega) = R(\omega)H(\omega)$.

12 The same filter on given numbers

Idea for this section

At one frequency, replace each bracket by the sum over the given records (the factor $1/(\text{number of records})$ cancels). Multiply X by the conjugate of R , not by R itself. If every X in the list is 0, the numerator is 0.

1. Two records at an in-band frequency have $(X, R) = (1, 2)$ and $(2, 4)$. Two records at an out-of-band frequency have $(X, R) = (0, 3)$ and $(0, -1)$. Find H at each frequency.
2. Two records at one frequency have $(X, R) = (i, 2)$ and $(1, i)$. Find H . Show the conjugate on each record before you add.

13 Three spectra on lined-up axes

Idea for this section

“Support” means where the magnitude is nonzero. Heights are not required. $|X|$ lives only where the input was given power. $|R|$ keeps going past that cutoff because spikes are deltas and deltas contain every frequency. $|\hat{X}| = |R||H|$ lives only where H is nonzero, which is the input’s support.

1. The input is white and band-limited to 6 Hz. On aligned frequency axes, sketch the qualitative support of $|X|$, $|R|$, and $|\hat{X}|$. Label the cutoffs in rad/s. In two sentences, say why $|R|$ is the wide one and why $|\hat{X}|$ is not.
2. The input is now a single 3 Hz cosine, not white noise. Sketch the same three supports. The input is no longer a filled rectangle. Say what replaces it, and what $|\hat{X}|$ then looks like.

14 Normalizing a PSC kernel

Idea for this section

The un-normalized piece is $t^n e^{-t/\tau}$ for $t \geq 0$. The constant c_n is its integral from 0 to ∞ , so $h_n = c_n^{-1} t^n e^{-t/\tau}$ has area 1. Here n is the integer power of t . It is not a frequency. For $n = 0$ the integral is immediate. For $n = 1$, integrate by parts; the boundary term vanishes and the remainder is τ times the $n = 0$ integral.

1. Find c_0 by evaluating the integral. Write $h_0(t)$ for $t \geq 0$, and its value at $t = 0$.
2. Find c_1 by integration by parts. Write $h_1(t)$. Give $h_1(0)$, the time of the peak, and the height of the peak. Set the derivative of $t e^{-t/\tau}$ to zero to locate the peak.

15 Drawing the two kernels

Idea for this section

h_0 jumps to $1/\tau$ at the spike and only decays. h_1 starts at 0 because of the factor t , and its peak is at $t = \tau$ with height $1/(e\tau)$. Both are zero for $t < 0$. For $n \geq 1$ the peak of the normalized kernel is at $t = n\tau$, and the value at zero is 0.

1. Sketch h_0 and h_1 on one pair of axes, at the same τ . Label both values at $t = 0$ and the time and height of each peak.
2. For $n = 2$, without repeating the integral: where is the peak, what is $h_2(0)$, and is this peak earlier or later than the peak of h_1 ?

16 A longer synaptic time constant

Idea for this section

For $n = 0$, $H(\omega) = 1/(1 + i\omega\tau)$ and the cutoff is $f_c = 1/(2\pi\tau)$. Increasing τ makes the unit-area kernel wider. Spikes are averaged over a longer interval, so the decoded trace loses rapid jitter. The same widening adds lag and pulls down the gain on fast parts of the signal. Residual spike noise is still there after filtering. Distortion is the lag and the attenuation of the signal itself.

1. With $n = 0$, give two changes in the decoded time trace when τ increases, and the mechanism for each.
2. One fixed $n = 0$ filter and one fixed decoder are used on two inputs of equal RMS, one band-limited to 2 Hz and one to 12 Hz. Both stay inside the trained range. Explain why the slower input is usually tracked more faithfully. Separate signal distortion from leftover spike noise. Three or four sentences.

17 The decoder, as a matrix

Idea for this section

A is $n \times N$: neurons by training samples. X is $d \times N$. The noise-regularized identity decoder is

$$D^T = (AA^T + N\sigma^2 I_n)^{-1} AX^T, \quad D \in \mathbb{R}^{d \times n}.$$

The matrix being inverted is $n \times n$. The penalty uses the sample count N , not the neuron count.

1. A population of n neurons represents a d -dimensional variable, with N training samples. Write D and state the shape of A , of X , of the matrix being inverted, and of D .
2. Now A is 40×1000 and X is 2×1000 , with $\sigma = 0.1$. What are n , d , and N ? What is inverted, and what is the shape of D ?

18 A decoder small enough to finish

Idea for this section

For one neuron, $AA^\top$ and $AX^\top$ are ordinary sums of products across the samples. The penalty $N\sigma^2$ is added to $AA^\top$ before dividing. Regularization makes $|d|$ smaller. The decoder has units of seconds when A is in hertz. RMSE is the square root of the average squared error on the training samples.

1. One neuron, two samples: activities $A = (2, 4)$, targets $X = (1, 2)$. Find the unregularized decoder. Then find the decoder whose penalty term $N\sigma^2$ equals 10. On the clean samples, which decoded values does each decoder produce?
2. One neuron, three samples: activities $(1, 1, 2)$, targets $(1, 1, 4)$, unregularized. Find d , the three decoded values, and the RMSE.

19 Spike counts and the factor of Δt

Idea for this section

Lecture 4 stores a discrete delta at height $1/\Delta t$ so that its area, height times Δt , equals 1. A bin count b_{ik} is already that area: 0 or 1, not a height. Samples h_k of a continuous unit-area PSC are in s^{-1} . The filtered rate is one copy of h per spike,

$$s_{ik} = \sum_j b_{ij} h_{k-j},$$

with $h_m = 0$ for $m < 0$. The Riemann sum $\sum (b_{ij}/\Delta t) h_{k-j} \Delta t$ cancels Δt , so the count formula does not contain another one. Then s is in hertz and the rate decoder applies directly.

1. Write s_{ik} as a sum over bins. State whether an extra Δt belongs in that sum, and connect the formula to the impulse-height convention in two sentences.
2. Spikes are stored as 1s, $\Delta t = 1$ ms, and someone normalizes the kernel samples so that they sum to 1 rather than using $h(t_k)$ in s^{-1} . A neuron fires steadily at 100 Hz. What value does this filtered train settle near, what value should a rate decoder see, and where can the missing factor be inserted? Name the three legal places.

20 From a function decoder to weights

Idea for this section

Population B encodes y with $J_i = \alpha_i \langle e_i, y \rangle + J_i^{\text{bias}}$. Substituting the estimate $\hat{y} = D^f a$ from population A makes the current a weighted sum of A's rates. The weight from pre-neuron j to post-neuron i is $w_{ij} = \alpha_i \langle e_i, d_j^f \rangle$, where d_j^f is column j of D^f . With row i of E equal to $\alpha_i e_i^\top$, one has $W^f = ED^f$. Bias is added once, outside W^f .

1. Population B has one neuron with $\alpha = 3$, unit encoder $e = +1$, and $J^{\text{bias}} = 1$. The function decoder from two presynaptic neurons is $d^f = (0.2, -0.1)$. Find both weights and write J in terms of the two rates a_1 and a_2 .

- Population A has $n = 10$ neurons. The decoded function is $q = 3$ dimensional. Population B has $m = 4$ neurons. State the shapes of D^f , E , and W^f .

21 A target of the form $2x + 1$

Idea for this section

The least-squares decoder is linear in the target. Decoding $af + bg$ uses $aD^f + bD^g$. In particular $D^{2x+1} = 2D + D^1$, where D^1 is the decoder whose target is the constant function 1. The decoder $2D$ alone reconstructs $2x$ and misses the offset.

- An identity decoder is $d = (0.4, -0.4)$. A decoder for the constant function 1 is $d^1 = (0.1, 0.1)$. Find the decoder for $2x + 1$. Why is $2d$ not that decoder?
- An identity decoder is $d = (1, -1)$ and $d^1 = (0.5, 0.5)$. Find the decoder for $-x + 2$.

22 Staying inside the trained range

Idea for this section

A population trained on an interval has decoders and tuning that were fitted there. If the target spends time outside that interval, the decoded output is an extrapolation: neurons saturate or sit near rates the decoders never saw, and the error grows. There is no requirement that the voltage or the rate hard-clips at the edge of the interval. If the whole target stays inside the interval, this particular limitation does not apply.

- Population B was trained on $[-1, 1]$. The connection implements $y = 2x + 1$ with $x(t) = \sin(2\pi t)$. Give the range of the desired y and the resulting limitation.
- The same population and the same $x(t)$ now implement $y = \frac{1}{2}x$. Give the range of y . Does the trained-range limitation apply?

23 Addition and multiplication

Idea for this section

Currents from separate populations add, so two incoming connections compute a function of x alone plus a function of y alone. A product xy is not of that form. Represent (x, y) together in one two-dimensional population, then decode the product. A nonlinearity of a single variable, such as y^2 , is a function decoder on that variable's own population. And $(x + y)^2$ can be computed by summing first, then squaring the scalar sum. Decoding x^2 and y^2 from the separate populations and adding those currents misses the cross term $2xy$.

- Draw the population-level network for $z = 4y - x$. External x and y begin as separate scalar populations. Label the dimension of every added population and the transformation on every connection.
- Draw the network for $p = xy$, with the same rule for labels.

3. Draw the network for $q = y^2$. Explain in one sentence why this one does not need a two-dimensional population.
4. Someone adds a decoded x^2 to a decoded y^2 and hopes to obtain $(x + y)^2$. Explain what is missing. Then draw a network that does compute $(x + y)^2$ from the two scalar populations.

24 The optimal filter in time

Idea for this section

The optimal $h(t)$ is nonzero for $t < 0$. Using it requires future spikes, so it is acausal: fine for offline analysis of a recording, and not a model of a synapse. Smoothing $H(\omega)$ by convolution against a window $W(\omega)$ multiplies $h(t)$ by a time window, which shortens and smooths h .

1. Say what “acausal” means for the optimal $h(t)$, why an offline decode may use it, and why a biological synapse cannot.
2. The windowed filter replaces $X\bar{R}$ and $|R|^2$ by their convolutions with $W(\omega)$. What does that operation do to $h(t)$, and why?

25 Three questions that each use two ideas

These are the first questions in the set that are allowed to feel like the practice test. Attempt all three before opening their solutions.

1. On $T = 1$ s the only nonzero coefficients are $X(\pm 2\pi) = 2 + 2i$, already conjugate-symmetric. Write $x(t)$ as a sine and a cosine. Find the RMS.
2. Both neurons use $J = ex + 1$ at the constant input $x = 0$. Describe both voltage traces. Do not assume that $J = 1$ produces spikes.
3. Draw the network for $z = y^2 + x$, starting from separate scalar populations for x and y . Say why a two-dimensional population is unnecessary here, whereas it was necessary for xy .

When every second question above can be done from a blank page, redo Practice Test 2 with the solutions to both papers closed. The original test is the same ideas with the hints removed.